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An Intelligent System for Dysarthria Classification of Male and Female Processed Dataset using Sequential Model Parameters

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Abstract— **There are several current studies including various methods for treating dysarthria in various developing countries. Research-based dysarthria treatments take a multidisciplinary approach that encompasses speech therapy, assistive technology, neurosurgery, pharmaceutical therapies, non-invasive brain stimulation, machine learning and AI methods. These methods seek to raise the general quality of life, communication and speaking abilities of dysarthria sufferers. With a 90% accuracy rate, the proposed sequential model was found to have good classification performance for identifying Dysarthria. Researchers have also been looking at the use of Machine Learning (ML) and Artificial Intelligence (AI) approaches to create tools and systems that can help people with dysarthria to speak and communicate for social welfare and provide access to best remedies.**

Keywords— ***CNN, Model Training, Dysarthria Classification, Sequential Model, Deep Learning, Confusion matrix***

I. INTRODUCTION

Several audio variables, including Mel-Frequency Cepstral Coefficients (MFCCs), Spectral Centroid, Spectral Contrast, Spectral Bandwidth, and Zero Crossing Rate are used to classify dysarthria using CNN on audio. These characteristics are taken from the audio source and sent into a CNN as classification input. The measures taken to achieve this were as follows:

- Audio data collection and pre-processing: Collect audio samples of speech with and without dysarthria, then use preprocessing to take out background noise and adjust loudness.
- Extract audio features: From the preprocessed audio data, use a feature extraction technique to extract the MFCCs, Spectral Centroid, Spectral Contrast, Spectral Bandwidth, and Zero Crossing Rate.
- Get the dataset ready: Create training, validation, and testing sets from the dataset. To guarantee that each set of speech samples contains an equal amount of dysarthric and non-dysarthric samples, the dataset must be balanced.
- Define the CNN model: Create an architecture for the CNN model that accepts audio information as input and produces the classification outcome. Convolutional, pooling, and fully linked layers can all be found in a CNN model.
- Develop the model: Utilise an appropriate optimizer, such as Adam or Stochastic Gradient Descent (SGD), to train the CNN model on the training dataset. In order to increase performance, keep an eye on the model's performance on the validation dataset and tweak hyperparameters like the learning rate, batch size, and number of epochs.
- Review the model: Measure the trained model's performance using the testing dataset and compute measures like accuracy, precision, recall, and F1-score.
- To increase performance, fine-tune the model by changing the architecture, the hyperparameters, or employing transfer learning strategies.

Overall, classifying dysarthria using CNN on audio is a difficult problem that necessitates a thorough grasp of feature extraction, signal processing, and deep learning. It is feasible to attain high accuracy in the categorization of dysarthria by selectively choosing and extracting pertinent audio data, as well as by utilising an appropriate CNN model architecture.

II. LITERATURE REVIEW

Convolutional Neural Networks (CNNs) were suggested by Yu, C. et al. to encode the motor information by including all speech function regions [1]. Using a Spatial Convolutional Neural Network (SCNN) and a multi-head attention transformer (MHAT) to visually extract speech data, Almadhor, A. et al. suggested a spatio-temporal dysarthric ASR (DASR)

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system. DASR then learns the forms of phonemes spoken by dysarthric people [2]. Using mel spectrograms, Joshy, A.A. et al. investigated the effectiveness of squeeze-and-excitation (SE) networks for categorising the severity degree of dysarthria [3]. An autonomous voice assistant system with speech recognition and auto-correction as its two primary components was introduced by Ali, S.K. et al [4]. Joshy, A.A. et al. investigated the efficacy of applying the multi-task learning strategy and the multi-head attention mechanism (MHA) on Dysarthria [5]. On the basis of 551,176 disordered speech samples provided by a broad group of 468 speakers who had a variety of self-reported speaking problems and were scored on a five-point scale for overall intelligibility, Venugopalan, S. et al. built dysarthric speech intelligibility classifiers [6]. Hybrid Acoustic Feature-Driven Deep Learning model for Dysarthric Speech Recognition and Perceptible Speech Generation was created by Latha, M. et al [7-8]. In order to objectively measure and categorise the pathological abnormalities in speech output, Liu, J. et al. provided a fresh viewpoint [9-10]. Hu, S. et al. developed multi-modal dysarthric and geriatric speech recognition systems using wav2vec2.0 representations in acoustic-to-articulatory (A2A) inversion [11-12]. For boosting the voice signal, Jolad, B. et al. suggested the Fractional Competitive Crow Search Algorithm-based voice Enhancement Generative Adversarial Network (FCCSA-SEGAN) [13-14]. The raw real and imaginary components of the Fourier transform of voice signals were used by Loweimi, E. et al. to examine a multi-stream acoustic modelling [15]. In order to translate encoded characteristics into intentions, Wang, P. et al. developed SLU systems that include a pre-trained speech representations encoder and an SLU decoder [16]. According to Karjigi, V. et al., Fisher vector encoding performs better than temporal encoding [17]. A Quantum Binary Classifier was developed by McComas, D., et al., and the outcomes were compared to the classical model [18-19]. A fine-tuned ResNet can successfully discriminate between stroke patients' appearance and voice, according to Lee, A. et al [20].

The results of a recent comprehensive study will be useful for future research on classification of Dysarthria. The primary conclusions of the study are summarised in the phrases that follow:

Throughout the whole investigation, Dysarthria Audio samples were categorised using the Sequential model.

• A variety of factors are investigated, including optimisation techniques and how they affect the results.

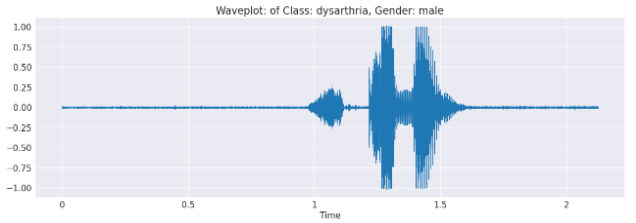
• Based on the criteria used in this study, this work may be beneficial to academics who have had difficulty coming up with a trustworthy method for detecting Dysarthria.

The following sections make up the study: Segments three and four present the proposed input dataset, segment five presents the data, and segment six concludes the presentation.

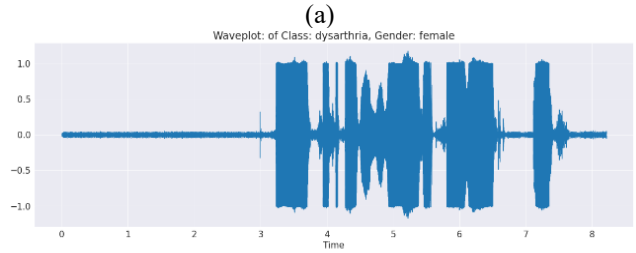
III. INPUT DATASET

The cerebral palsy (CP) and amyotrophic lateral sclerosis (ALS), two of the most common causes of speech disability (Kent and Rosen, 2004), and matched controls' aligned acoustics and measured 3D articulatory features make up the TORGO database of dysarthric articulation. The Holland-Bloorview Kids Rehab facility in Toronto and the Computer Science and Speech-Language Pathology departments at the University of Toronto collaborated to create this database, known as TORGO.

This dataset includes 2000 samples of males and females with dysarthria, as well as males and females without the condition.



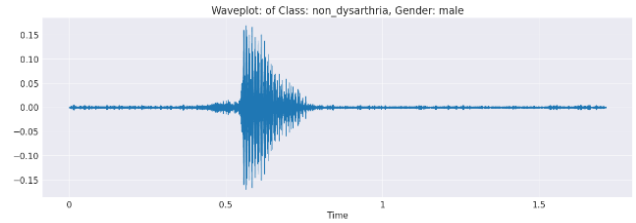
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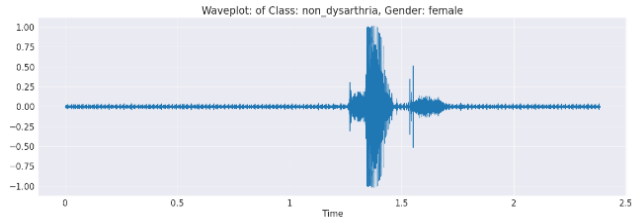
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Fig. 1 Waveplot of Dysarthria affected (a) Male (b) Female Voice

The male sample from the dysarthric samples, which is shown in Fig. 1 and Fig. 2, has slurred speech, as shown by the waveplot, while the female sample's speech is quite rapid and difficult to understand because the words overlap, as shown by the box-like patterns that appear in the waveplot. The non-dysarthric samples exhibit a regular waveplot, which suggests that the speech is regularly timed.



(a)



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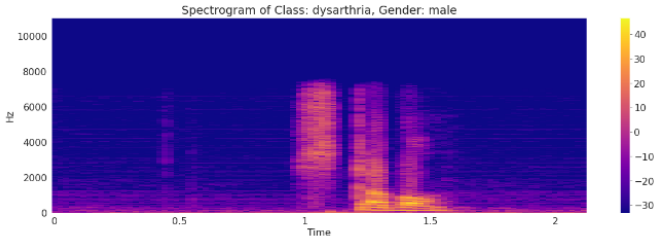
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Fig. 2 Wave plot of Non- Dysarthria affected (a) Male (b) Female Voice

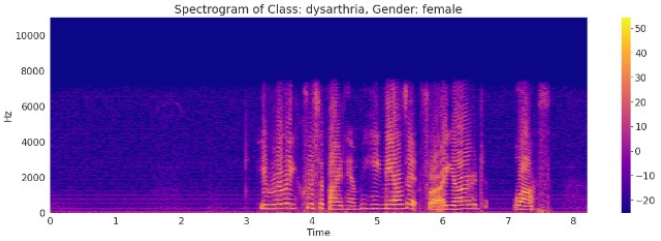
IV. DYSARTHRIA CLASSIFICATION USING CNN ON AUDIO USING VARIOUS PARAMETERS

A. *Evaluation of Dysarthria using Spectrograms*

It is evident from the dysarthric samples as shown in Fig. 3 and Fig. 4 that the energy magnitudes of the frequencies are more evenly distributed throughout time, a sign of sluggish, slurred speech or because the syllables are being said more quickly and overlapping one another.



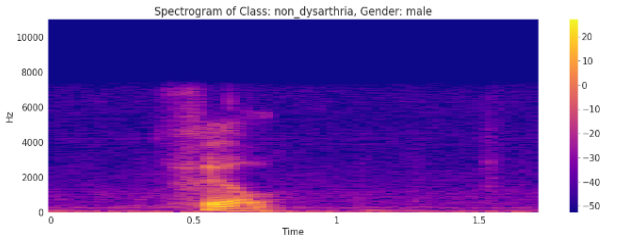
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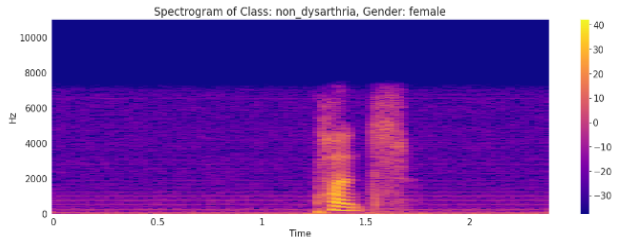
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Fig. 3 Spectrogram of Dysarthria affected (a) Male (b) Female Voice

Similar patterns could also be seen in monotonous, dysarthric speech. The energy magnitudes for the non-dysarthric samples may be seen to be more restricted to only the areas when they have uttered anything, which is regular paced.



(a)

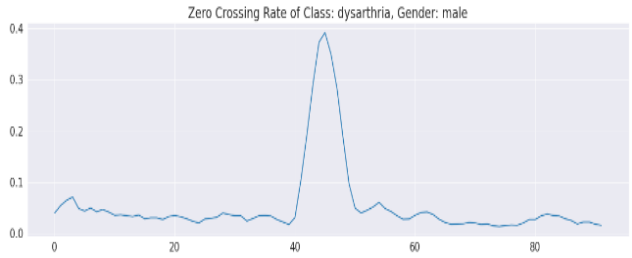


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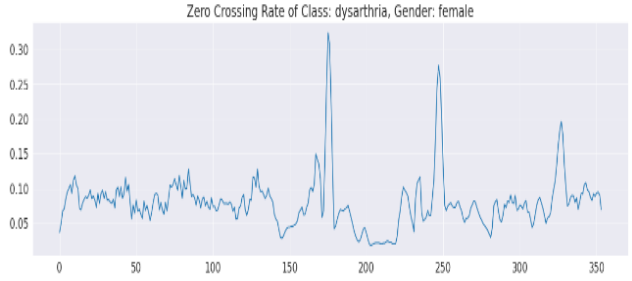
Fig. 4 Spectrogram of Non- Dysarthria affected (a) Male (b) Female Voice

B. *Calculation of Zero Crossing Rate*

For dysarthric samples as shown in Fig.5 and Fig. 6, it can be seen that there were times when some of their words were substantially louder in the audio itself (for the female dysarthric case, for example), leading to several peaks of zero crossing rate in those situations. Their inability to control their tongue and other mouth-related muscles when speaking and their inability to purposefully emphasis words may be the cause of the abrupt loudness of some phrases.



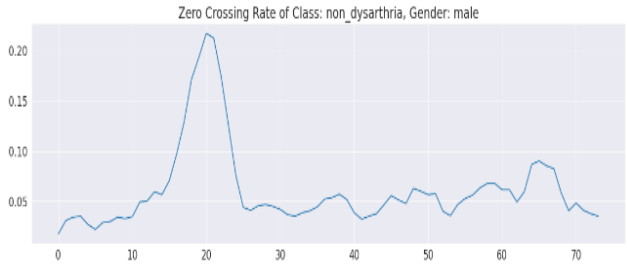
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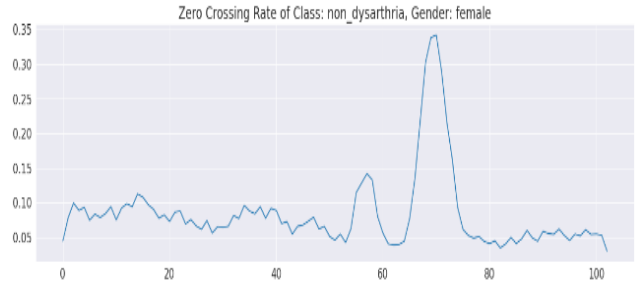
Fig. 5 Zero Crossing Rate for Dysarthria of (a) Male (b) Female Voice

In scenarios where the speaker's voice can be heard, the zero crossing rate for non-dysarthric individuals is significant.



(a)

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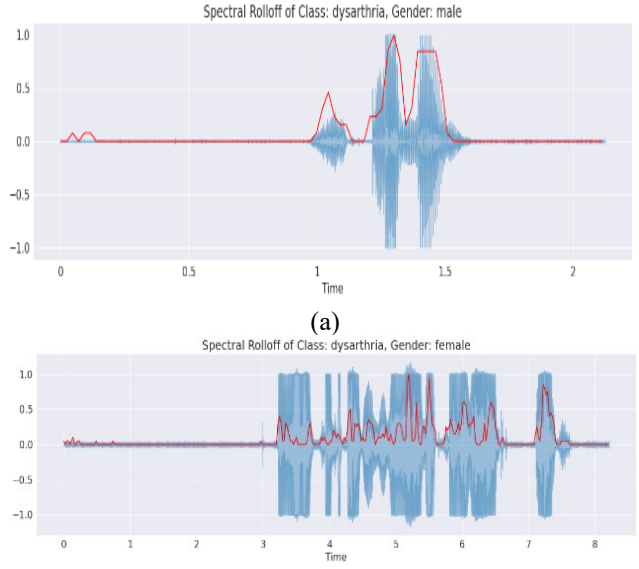
(b)

Fig. 6 Zero Crossing Rate for Non-Dysarthria of (a) Male (b) Female Voice

C. *Indication of Frequency using Spectral Rolloff*

The frequency below which a certain percentage (librosa: 85%) of the magnitude distribution is concentrated is shown by spectral rolloffs in Fig. 7 and Fig. 8.

The 85% coverage below the rolloff frequency can be seen surrounding the speech present in the audio signal, which is consistent with our earlier observations in the spectrograms that the energy magnitudes of the frequencies are more dispersed in the case of dysarthric samples.

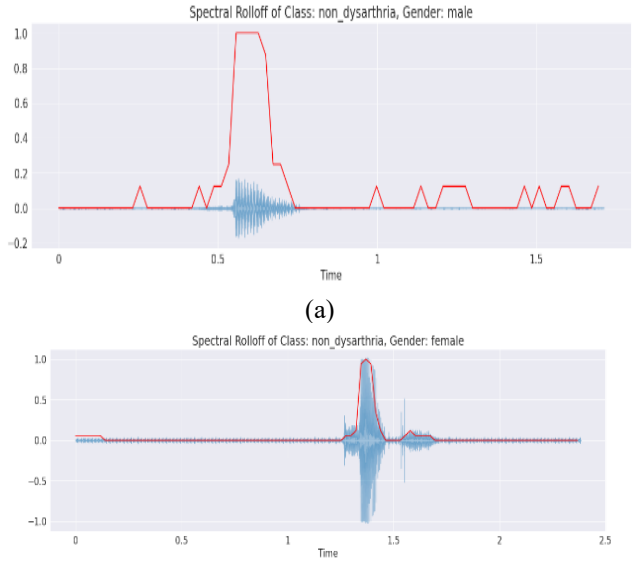


(a)

(b)

Fig. 7 Spectral Rolloff for Dysarthria of (a) Male (b) Female

For non-dysarthric samples in Fig. 8, we can observe that the 85% coverage includes both the speech and some additional audio stutters and noise.



(a)

(b)

Fig. 8 Spectral Roll off for Non-Dysarthria of (a) Male (b) Female

D. *Dysarthria Diagnosis using Feature Extraction*

Various signal processing methods, including spectrum analysis, autocorrelation, and cepstral analysis, can be used for feature extraction. These characteristics can be used to train machine learning algorithms to create models for dysarthria diagnosis and monitoring. Due to nerve injury, dysarthria is a speech condition that impairs the capacity to pronounce sounds, words, and sentences. Analysing the speech signal to extract pertinent features that may be utilised to detect and track the condition is the process of feature extraction for dysarthria.

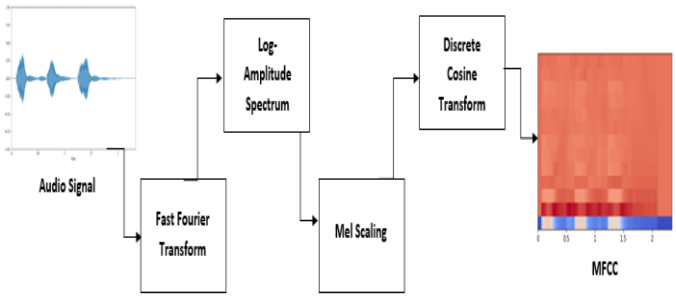


Fig. 9 Feature extraction algorithm

This study extracted 128 MFCC Features, which was the maximum number of features that could be extracted during the extraction procedure, as shown in Fig. 9.

V. RESULTS

A. *Modelling of Input images on Sequential Model*

Using a neural network architecture that is made to handle sequential data, such as time-series data, text, or pictures, is necessary for modelling input images on a sequential model. A sequential model as shown in Fig. 10 can be used to image

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data to analyse pictures as a series of pixels or as a series of smaller image patches.

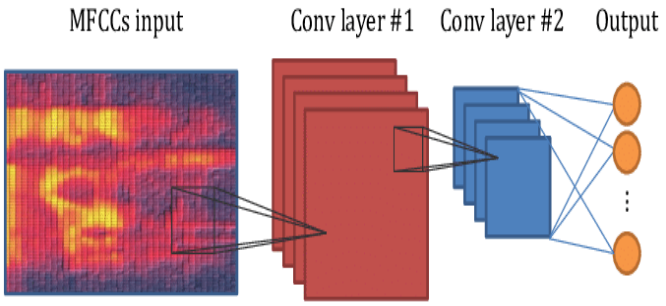


Fig. 10 Modelling input images of Sequential model

Convolutional Neural Networks (CNNs), a kind of neural networks made expressly for handling image data, are one method for modelling input pictures on a sequential basis. Convolutional layers and pooling layers are used by CNNs to extract features from the input pictures and to compress the feature maps as shown in Fig. 11.

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 16, 8, 32)	320
max_pooling2d (MaxPooling2D)	(None, 8, 4, 32)	0
conv2d_1 (Conv2D)	(None, 8, 4, 64)	18496
max_pooling2d_1 (MaxPooling2D)	(None, 4, 2, 64)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 32)	16416
dense_1 (Dense)	(None, 1)	33

Total params: 35,265
Trainable params: 35,265
Non-trainable params: 0

Fig. 11 Layer wise depiction of Sequential Model

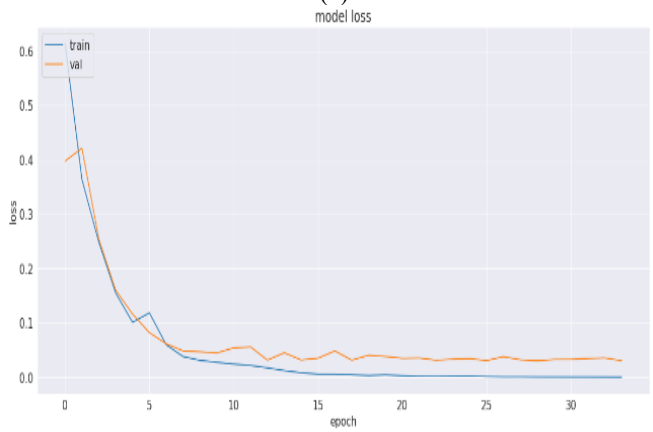
Epoch	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
1	0.6137	0.7154	0.3977	0.8100
5	0.1012	0.9722	0.1164	0.9700
10	0.0277	0.9944	0.0448	0.9850
15	0.0059	1.0000	0.0351	0.9900
20	0.0030	1.0000	0.0350	0.9850
25	0.0015	1.0000	0.0309	0.9900
30	0.0012	1.0000	0.0329	0.9900

Fig. 12 Epoch Depiction on Adam Optimizer

The complexity of the model as shown in Fig. 12, the size of the training dataset, and the learning rate are some of the variables that affect how many epochs are needed to train a sequential model. It is customary to stop training a model when its performance on a validation dataset begins to plateau, signalling that more training is unlikely to yield better results.



(a)



(b)

Fig. 13 Training and Validation (a) Accuracy (b) Loss on visualized curves

The validation accuracy is seen changing over time in Fig. 13 (a) throughout the training phase of the model. Additionally, as the model develops and gets more precise, it should grow larger. The validation accuracy is used to monitor the generalizability of the model.
Fig. 13 (b) shows a time-varying representation of the training loss during the model's training phase. It should decrease over time as the model matures and gets more precise.

B. *Comparison of the Confusion Matrix Parameters*

An effective method for evaluating how well a machine learning model for Dysarthria analysis is doing is a confusion matrix, such as the one in Fig. 14. It provides an overview of the existing and anticipated labels used in the categorization operation. The confusion matrix may be used to construct performance metrics like as accuracy, precision, recall, and F1 score, among others, to provide a more detailed evaluation of the model's performance.

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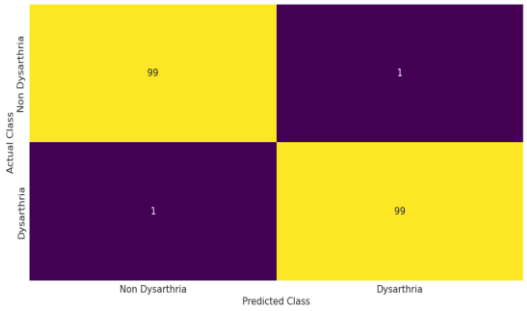


Fig. 14 Classification of Dysarthria using the Confusion Matrix Parameters

VI. CONCLUSION

Dysarthria can be caused by a number of conditions, including brain injuries, strokes, and drugs. Neurological conditions including Parkinson's disease, multiple sclerosis, and ALS are among the causes of dysarthria. Speech therapy, assistive technology, medication, and surgical procedures are all possible forms of treatment for dysarthria. For those with dysarthria, therapy aims to enhance speech production, communication skills, and quality of life. Machine Learning (ML) and Artificial Intelligence (AI) approaches have showed potential in treating dysarthria. AI and ML methods have the potential to offer efficient treatments for dysarthria, enhancing the quality of life and communication for those who suffer from the disorder. This study shows that the suggested Sequential model can classify Dysarthria from processed audio samples with 99% accuracy. Overall, prior studies on dysarthria have concentrated on a number of facets of the disorder, such as diagnosis, evaluation, and therapy, as well as the connection between acoustic characteristics and perceived dysarthria severity.

REFERENCES

[1] Yu, C., Su, X. and Qian, Z., 2023. Multi-stage Audio-Visual Fusion for Dysarthric Speech Recognition with Pre-trained Models. IEEE Transactions on Neural Systems and Rehabilitation Engineering.

[2] Almadhor, A., Irfan, R., Gao, J., Saleem, N., Rauf, H.T. and Kadry, S., 2023. E2E-DASR: End-to-end deep learning-based dysarthric automatic speech recognition. Expert Systems with Applications, 222, p.119797.

[3] Joshy, A.A. and Rajan, R., 2023. Dysarthria severity assessment using squeeze-and-excitation networks. Biomedical Signal Processing and Control, 82, p.104606.

[4] Ali, S.K., Hassan, A.M., Salah, A.M., Mohamed, D.W., Mustafa, M. and Mohamed, E.H., 2023. Re-Talk: Automated Speech Assistance for People with Dysarthria.)1(5, النشرة المعلوماتية في الحاسبات والمعلومات, pp.16-22.

[5] Joshy, A.A. and Rajan, R., 2023. Dysarthria severity classification using multi-head attention and multi-task learning. Speech Communication, 147, pp.1-11.

[6] Venugopalan, S., Tobin, J., Yang, S.J., Seaver, K., Cave, R.J., Jiang, P.P., Zeghidour, N., Heywood, R., Green, J. and Brenner, M.P., 2023. Speech Intelligibility Classifiers from 550k Disordered Speech Samples. arXiv preprint arXiv:2303.07533.

[7] Latha, M., Shivakumar, M., Manjula, G., Hemakumar, M. and Kumar, M.K., 2023. Deep Learning-Based Acoustic Feature Representations for Dysarthric Speech Recognition. SN Computer Science, 4(3), p.272.

[8] Gill, K.S., Anand, V., Gupta, S. and Figat, P., 2023, February. Stratification of White Blood Cells Using Optimized DenseNet201 Model. In Advanced Communication and Intelligent Systems: First International Conference, ICACIS 2022, Virtual Event, October 20-21, 2022, Revised Selected Papers (pp. 31-47). Cham: Springer Nature Switzerland.

[9] Liu, J., Du, X., Lu, S., Zhang, Y.M., An-ming, H.U., Ng, M.L., Su, R., Wang, L. and Yan, N., 2023. Audio-video database from subacute stroke patients for dysarthric speech intelligence assessment and preliminary analysis. Biomedical Signal Processing and Control, 79, p.104161.

[10] Gill, K.S., Sharma, A., Anand, V. and Gupta, R., 2022, December. Brain Tumor Detection using VGG19 model on Adadelta and SGD Optimizer. In 2022 6th International Conference on Electronics, Communication and Aerospace Technology (pp. 1407-1412). IEEE.

[11] Anand, V., Gupta, S., Altameem, A., Nayak, S.R., Poonia, R.C. and Saudagar, A.K.J., 2022. An Enhanced Transfer Learning Based Classification for Diagnosis of Skin Cancer. Diagnostics, 12(7), p.1628.

[12] Hu, S., Xie, X., Jin, Z., Geng, M., Wang, Y., Cui, M., Deng, J., Liu, X. and Meng, H., 2023. Exploring Self-supervised Pre-trained ASR Models For Dysarthric and Elderly Speech Recognition. arXiv preprint arXiv:2302.14564. [13] Prema, C.E., Suresh, S., Krishnan, M.N. and Leema, N., 2022. A Novel Efficient Video Smoke Detection Algorithm Using Co-occurrence of Local Binary Pattern Variants. Fire Technology, 58(5), pp.3139-3165.

[14] Jolad, B. and Khanai, R., 2023. An approach for speech enhancement with dysarthric speech recognition using optimization based machine learning frameworks. International Journal Of Speech Technology, pp.1-19.

[15] Loweimi, E., Yue, Z., Bell, P., Renals, S. and Cvetkovic, Z., 2023. Multi-Stream Acoustic Modelling Using Raw Real and Imaginary Parts of the Fourier Transform. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31, pp.876-890.

[16] Wang, P., 2023. Benefits of pre-trained mono-and cross-lingual speech representations for spoken language understanding of Dutch dysarthric speech. EURASIP Journal on Audio, Speech, and Music Processing, 2023(1), pp.1-25.

[17] Karjigi, V. and Sreedevi, N., 2023. Speech intelligibility assessment of dysarthria using Fisher vector encoding. Computer Speech & Language, 77, p.101411.

[18] McComas, D., REU Project: Quantum Machine Learning for Dysarthric Speech Detection.

[19] Gundaboina, L., Badotra, S., Bhatia, T.K., Sharma, K., Mehmood, G., Fayaz, M. and Khan, I.U., 2022. Mining cryptocurrency-based security using renewable energy as source. Security and Communication Networks, 2022. [20] Lee, A., Che, D., Qi, K., Patton, E. and Zeng, T., 2023, February. Mobile device application of deep learning for rapid onset stroke detection. In AIP Conference Proceedings (Vol. 2562, No. 1, p. 020006). AIP Publishing LLC.

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